

RentWise – AI Based Risk-Aware Peer-to-Peer Rental Management Platform



Group members

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DECLARATION

I certify that project title RentWise is under my supervision with students of BSCS-23, Faculty of Computing & Information Technology, University of Gujrat, Pakistan, worked under my supervision.

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Dated: September 16, 2026

FINAL YEAR PROJECT PROPOSAL

Abstract

Peer-to-peer rental platforms allow individuals to access expensive assets without purchasing them, but transactions between strangers can create concerns about identity, asset condition, late returns, disputes, and financial risk. RentWise proposes a risk-aware rental management platform for high-value assets such as cameras, laptops, projectors, gaming equipment, and professional tools. The system will provide a Flutter mobile application for renters and owners and a Next.js web application for owners and administrators, supported by Supabase for authentication, database, storage, and security. Core modules include asset listing, discovery, booking, verification status, rental agreements, condition evidence, returns, ratings, and dispute management. Its AI layer will include machine-learning-based transaction risk prediction with explainable factors, personalized asset recommendation using natural-language requirements, and an AI Rental Assistant. NLP-based review/dispute analysis and computer-vision damage detection will remain optional subject to time and resources. The project will use Flutter, Next.js, Supabase, Python, machine-learning libraries, LLM/NLP services, GitHub, and Figma. Future work can extend the platform with additional intelligent rental services.

1. Introduction

RentWise is a peer-to-peer rental management platform designed to improve the management and trust of high-value asset rentals. Existing rental and classified platforms mainly support listing and discovery, while users may still need to assess reliability, asset condition, rental risk, and disputes manually. RentWise combines a structured rental lifecycle with verification status, reputation, evidence management, AI-based risk assessment, recommendations, and natural-language assistance.

2. Project Title

RentWise: AI Based Risk-Aware Peer-to-Peer Rental Management Platform

3. Project Overview

RentWise will provide an integrated rental lifecycle for high-value personal assets. The mobile application will primarily support renters and owners with asset discovery, rental requests, booking, agreements, handover, returns, evidence, ratings, and AI assistance. The web application will primarily support owners and administrators through listing management, rental monitoring, risk information, dispute handling, and analytics. Supabase will provide authentication, PostgreSQL database services, storage, and security. The AI layer will provide transaction risk prediction, explainable risk information, personalized asset recommendations, and an AI Rental Assistant. Optional NLP and computer-vision modules will be considered if sufficient time and resources remain.

4. Problem Statement

Existing rental and classified platforms mainly focus on listing and discovering assets. However, users may still have to manually assess another party's reliability, evaluate transaction risk, maintain condition evidence, and manage disputes.

RentWise aims to provide a structured rental lifecycle with AI-assisted risk assessment, recommendations, natural-language assistance, verification status, evidence management, and dispute handling within a single platform.

5. Targeted Audience

- **Renters:** students, freelancers, content creators, event organizers, professionals, and individuals needing high-value equipment temporarily.
- **Asset owners:** individuals, freelancers/professionals, and small rental businesses with cameras, laptops, projectors, gaming equipment, tools, and similar assets.
- **Administrators:** users responsible for verification, listings, transactions, disputes, risk flags, and platform monitoring.

6. Project Goals & Objectives

6.1 Goal

To develop an AI-enhanced peer-to-peer rental platform that enables users to discover, rent, manage, return, and review high-value assets while providing intelligent risk assessment, recommendations, and rental assistance.

6.2 Objectives

- Develop a cross-platform Flutter mobile application for renters and owners.
- Develop a responsive Next.js web application for owners and administrators.
- Implement secure authentication and role-based user management.
- Implement asset listing, discovery, availability, rental requests, and booking.
- Develop digital rental agreements and structured handover/return workflows.
- Implement verification status, reputation, ratings, and transaction history.
- Maintain pre-rental and post-rental condition evidence.
- Develop an ML-based transaction risk prediction module.
- Provide understandable factors contributing to predicted transaction risk.
- Develop personalized asset recommendations using user requirements and available asset information.
- Integrate an AI Rental Assistant capable of understanding natural-language rental queries.
- Implement dispute management with administrator-controlled decisions.
- Evaluate the functionality, usability, and intelligent features of the developed system.

7. Main System Modules

7.1 User Authentication & Profiles

Manages registration, login, user roles, profiles, verification status, rental history, and reputation.

7.2 Asset Listing & Management

Allows owners to add assets, upload images, provide specifications, set rental prices and availability, and maintain condition records. The initial focus will be on high-value assets such as cameras, laptops, projectors, gaming equipment, and professional tools.

7.3 Asset Search & Discovery

Enables renters to search and filter assets based on category, price, availability, specifications, owner reputation, and condition.

7.4 Rental Request & Booking

Manages rental requests, owner decisions, booking status, rental dates, availability, and transaction history.

7.5 Rental Agreement & Transaction

Provides structured digital rental records containing rental terms, duration, price, deposit information, handover and return requirements, and agreement status.

7.6 Verification & Trust

Maintains verification status, account history, completed rentals, ratings, late returns, and dispute history. Only necessary verification information will be stored.

7.7 Evidence & Condition Management

Stores pre- and post-rental images, condition descriptions, and supporting evidence associated with rental transactions.

7.8 Transaction Risk Assessment

Evaluates rental transactions using factors such as asset value, rental duration, verification status, user history, ratings, late returns, and disputes to provide a risk level and supporting factors.

8. AI Modules

8.1 AI Transaction Risk Prediction — Core

Uses machine-learning models such as Random Forest or XGBoost to predict the risk associated with a rental transaction based on user, asset, and transaction-related factors.

8.2 Explainable AI Risk Assessment — Core

Provides understandable contributing factors behind the predicted risk score, allowing owners and administrators to make informed decisions.

8.3 Personalized Asset Recommendation — Secondary

Uses natural-language requirements, asset specifications, price, availability, and user preferences to recommend suitable rental assets.

8.4 AI Rental Assistant — Secondary

Provides a natural-language interface for asset search, availability queries, rental guidance, recommendations, risk explanations, and agreement/return information through controlled system functions.

8.5 NLP Review & Dispute Analysis — Optional

Analyzes reviews and dispute descriptions for sentiment, complaint categories, and issues requiring administrator attention. Final decisions remain human-controlled.

8.6 Computer Vision Damage Detection — Optional

May compare pre- and post-rental asset images to identify possible damage or changes. The feature will assist evidence analysis and will not automatically assign liability.

9. Application Architecture

9.1 Web Application

The web application will primarily support owners and administrators.

9.1.1 Owner Dashboard

Provides listing and availability management, rental request handling, renter verification and risk information, agreements, active rentals, condition evidence, returns, and ratings.

9.1.2 Admin Dashboard

Provides user and listing management, verification review, transaction monitoring, risk-flag monitoring, evidence and dispute management, and system activity monitoring.

9.2 Mobile Application

The Flutter mobile application will primarily support renters and owners.

9.2.1 Renter Features

Registration, profile management, asset search and filtering, asset details, recommendations, AI Rental Assistant, rental requests, booking, agreements, handover/return, evidence, ratings, and disputes.

9.2.2 Owner Features

Profile management, asset listing and availability, rental requests, risk information, agreements, handover/return, condition evidence, ratings, and rental history.

9.3 Feature Prioritization

Core / Must Implement

- Mobile application
- Web application
- Backend and database
- Authentication and profiles
- Asset management
- Search and discovery
- Rental requests and booking
- Rental agreements
- Verification and trust
- Evidence management

- Returns, ratings, and disputes
- ML-based risk prediction
- Explainable risk assessment

Secondary Priority

- Personalized asset recommendation
- AI Rental Assistant

Optional

- NLP review and dispute analysis
- Computer vision damage detection
- Advanced semantic search

This priority-based approach ensures that optional AI features do not compromise the completion of the core rental management and risk-assessment system.

RentWise AI - High-Level Application Architecture

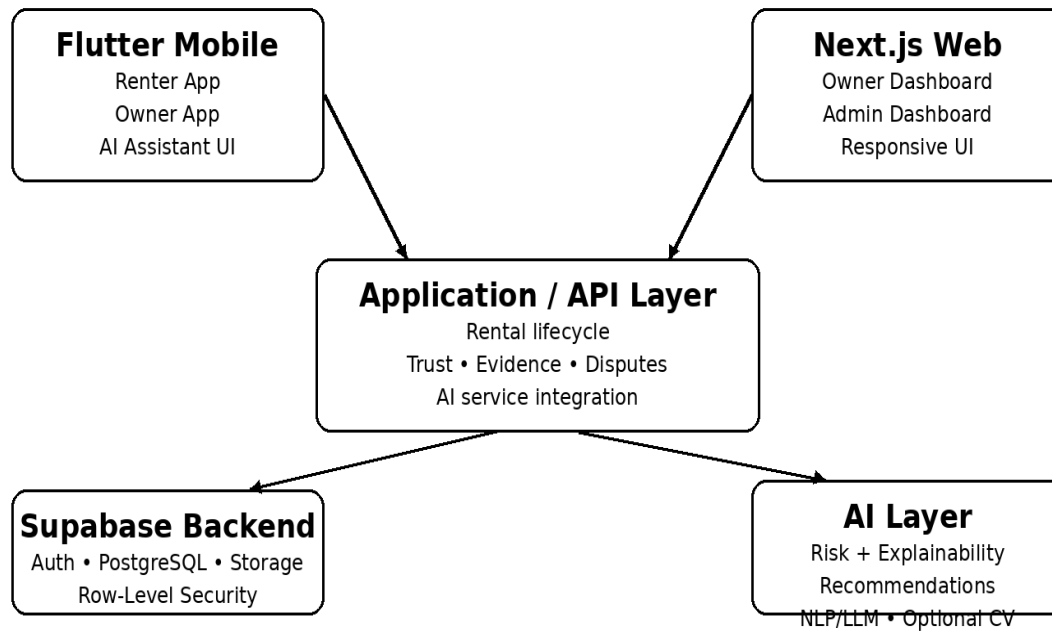


Figure 1

10. Competitive Analysis

Feature	RentWise	Typical Rental/Classified Platforms
Rental listings	✓	✓
High-value asset focus	✓	Partial
Verification status	✓	Limited/varies
ML transaction risk prediction	✓	Generally not core
Explainable risk factors	✓	Generally not core
AI recommendations	✓	Limited/varies
AI rental assistant	✓	Generally not core
Condition evidence	✓	Limited/varies
Rental agreements	✓	Varies
Return workflow	✓	Varies
Evidence-based disputes	✓	Varies
Admin risk monitoring	✓	Varies

11. Tools and technologies used with reasoning

Technology	Reasoning
Flutter / Dart	One codebase for Android/iOS and suitable for the mobile rental workflows.
Next.js / React	Responsive web interfaces for owner and admin dashboards.
Supabase	PostgreSQL, authentication, storage and row-level security with low setup overhead.
Python	Suitable for machine learning, model evaluation and AI services.
Scikit-learn / XGBoost	Established algorithms for classification-based transaction risk prediction.
Explainability techniques	Present understandable factors behind risk predictions.
LLM / NLP services	Natural-language requirements, rental assistance and selected language processing.
Git / GitHub	Version control, branching and collaboration.
Figma	Shared UI/UX design, prototyping and consistency.

12. Project milestones and deliverables

Milestone	Main Deliverables
1. Requirements & Planning	Requirements, use cases, architecture, initial backlog
2. UI/UX Design	Figma prototypes and mobile/web screen designs
3. Backend Foundation	Database schema, authentication, roles, storage, security
4. Core Mobile Development	Renter/owner flows, listings, search, booking, rental lifecycle
5. Web Development	Owner and admin dashboards
6. Core AI	Risk prediction, evaluation and explainable output
7. Secondary AI	Recommendations and AI Rental Assistant
8. Optional AI	NLP review/dispute analysis and/or limited CV damage detection
9. Testing & Integration	Testing, bug fixing and usability evaluation
10. Finalization	Deployment, documentation, presentation and final demonstration

13. Estimated Cost

The project will primarily use free and open-source technologies and available free-tier services. The final cost will depend on AI API usage, hosting requirements, domain selection, and whether optional paid services are used. Development will initially prioritize free-tier and open-source resources.

14. Work division among Group members

Member	Role	Responsibilities
Areeba Arif	Core Developer, Mobile & AI Engineer	Flutter architecture and mobile app; core rental workflows; ML risk prediction and explainability; recommendation; AI assistant; AI evaluation; technical integration.
Noor Fatima	UI/UX Designer & Frontend Developer	UI/UX research; Figma; mobile UI; Next.js web app; owner dashboard; responsive design; frontend integration; usability testing and AI feature interfaces.

Imtishal Abid	Backend, Admin & Quality Engineer	Supabase schema; authentication/authorization; storage/security/RLS; admin dashboard; backend transaction services; evidence/disputes; integration/system testing; deployment and documentation support.
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